



Review Paper

Machine learning for predicting chronic diseases: a systematic review

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ABSTRACT

Objectives: We aimed to review the literature regarding the use of machine learning to predict chronic diseases.

Study design: This was a systematic review.

Methods: The searches included five databases. We included studies that evaluated the prediction of chronic diseases using machine learning models and reported the area under the receiver operating characteristic curve values. The Transparent Reporting of a multivariable prediction model for Individual Prognosis Or Diagnosis scale was used to assess the quality of studies.

Results: In total, 42 studies were selected. The best reported area under the receiver operating characteristic curve value was 1, whereas the worst was 0.74. K-nearest neighbors, Naive Bayes, deep neural networks, and random forest were the machine learning models most frequently used for achieving the best performance.

Conclusion: We found that machine learning can predict the occurrence of individual chronic diseases, progression, and their determinants and in many contexts. The findings are original and relevant to improve clinical decisions and the organization of health care facilities.

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Introduction

The demographic transition is increasing the presence of chronic diseases around the world, especially in low- and middle-income countries. At least one in three adults has multiple chronic diseases, also called multimorbidity, associated with higher poor outcomes and health care costs.¹ In the United States, the direct costs of chronic diseases for the health care systems are nearly US\$ 214 billion a year. Furthermore, chronic diseases also cost US\$ 138 billion in lost productivity at work.² Multimorbidity is associated with functional disability and frailty, especially in older adults.³ Moreover, a review study showed that multimorbidity predicts future functional decline with greater impact in those with a higher number of diseases.⁴

Quality of life decreases with higher multimorbidity, mainly because of its effect on physical health.⁵ According to a study of

elderly patients, the perception of the quality of life is affected by multimorbidity individuals, mainly regarding physical and psychological conditions.⁶ Solutions based on artificial intelligence (e.g. machine learning) can support decision-making about the application of resources, ensuring efficiency and cost-effectiveness in the health care system. Machine learning algorithms are able to support intelligent decisions by using different data types (demographic, laboratory, and image data) to make predictions of disease risk, diagnosis, prognosis, and appropriate treatments.⁷

Machine learning is based on the idea that computers can learn to perform certain activities from data without manual programming.⁸ These methods are based on computational algorithms that relate all or some elements from a set of predictor variables to an outcome. The models depend on the type of variables that are available for the learning phase. If the outcome of interest is a quantitative variable, it is known as a regression model, whereas for qualitative or categorical tasks, it is known as a classification model.⁸

Despite its recent growth in health care, some studies have used machine learning algorithms to establish the diagnosis and treatment of diseases, managing to assess with reasonable precision the

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incidence of some diseases, such as diabetes,^{9,10} hypertension,¹¹ and chronic kidney disease.¹² Several metrics can be used to evaluate the performance of a predictive algorithm, with the area under the receiver operating characteristic curve (AUC) being the most common. The AUC value ranges from 0 to 1, so that the closer to 1, the better the performance.^{13,14}

Given the growing occurrence of chronic diseases¹⁵ and the potential of machine learning to improve decisions regarding these conditions, this study aimed to review the literature that analyzed the performance of machine learning algorithms in predicting chronic non-communicable diseases.

Methods

We conducted a systematic review following the recommendations of the Preferred Reporting Items for Systematic Reviews and Meta-Analyses.¹⁶ The guiding question of the review was as follows: Which studies have been published regarding the performance of machine learning models in predicting chronic diseases? This study was also registered on the International Prospective Register of Systematic Reviews (CRD42020215581).

Inclusion criteria

We included studies that evaluated the prediction of chronic non-communicable diseases using machine learning algorithms (models) and presented the results of the AUC.

Exclusion criteria

We excluded review studies and those that did not involve human beings.

Search strategy

We performed the search until April 2021 in the following databases: PubMed, LILACS, SciELO, Web of Science, and Scopus. Restrictions regarding the language or year of publication were not applied. We used two groups of keywords to search articles selected based on the Medical Subject Headings system. In the first group, we used the term: 'machine learning.' In the second, we considered the following terms for chronic diseases: 'chronic diseases,' 'disease, chronic,' 'diseases, chronic,' 'heart attack,' 'cardiovascular diseases,' 'cancer,' 'chronic obstructive pulmonary disease,' and 'type 2 diabetes.' Within each group, the Boolean operator 'OR' was used, and between groups, the Boolean operator 'AND' was used.

Study selection

Initially, a database in the Endnote X7 program was generated from the searches performed. The selection of articles included in the review was carried out individually by two reviewers (F.M.D. and A.K.C.) based on the previously defined inclusion and exclusion criteria. In case of disagreement, the two authors discussed to establish a consensus. Initially, each reviewer independently selected the titles of the articles of interest. The second stage consisted of reading the abstracts, followed by a full reading of the selected papers. Finally, the references of these articles were revised.

Risk of bias

There are no quality scales for machine learning predictions. We used an adapted version of Transparent Reporting of a

multivariable prediction model for Individual Prognosis Or Diagnosis (TRIPOD) with 20 items (14 for methods and six for results, Supplementary material 2).^{17,18} TRIPOD includes a checklist, in which the aim is to improve the reporting of studies that develop, validate, or update a prediction model for diagnostic or prognostic purposes.¹⁸

Results

Characteristics of included studies

Fig. 1 shows the flowchart of the study selection process. After the exclusion of duplicated articles, we found 1891 titles. We selected 245 abstracts for reading, from which we identified 124 articles for a full reading. The final sample was composed of the 42 articles included in the review. The reasons for exclusions of the 82 studies were review studies ($n = 3$), did not predict chronic diseases ($n = 50$), did not show the AUC values ($n = 22$), complete text was not found ($n = 5$), and repeated study ($n = 2$).

Thirteen studies analyzed data from China,^{19–31} five from India,^{32–36} five from the United States,^{37–41} two from Brazil,^{42,43} two from Canada,^{9,44} two from Finland,^{45,46} two from Taiwan,^{12,47} one from Oman,⁴⁸ one from Iraq,⁴⁹ one from Pakistan,⁵⁰ one from Egypt,⁵¹ one from Switzerland,⁵² one from Japan,⁵³ one from Italy,⁵⁴ one from Germany,⁵⁵ one from Australia,⁵⁶ and one from Jordan.⁵⁷ Moreover, one study combined databases from Italy and the United States.⁵⁸ The number of total individuals evaluated ranged from 50 (in two studies)^{38,42} to 680,810.³⁷ The most evaluated diseases were diabetes^{9,19,23,30,31,34,36,38–40,43,48,56} and chronic kidney disease,^{12,33,35,49,52,55,59} with 13 and seven studies, respectively. Support vector machine ($n = 22$), random forest ($n = 17$), and logistic regression ($n = 16$) were the machine learning models most used by the studies. Around 74% of the studies were published in the last 2 years.

Main findings

The amplitude of the AUC values ranged from 0.74 in a study that evaluated the prediction of cardiovascular events in 1216 individuals⁵⁸ to 1 in studies that predicted chronic kidney disease in 224 individuals,³⁵ evaluated chronic obstructive pulmonary disease in 50 individuals,⁴² predicted diabetes in 952 individuals,³⁶ and predicted lung cancer in 952 individuals from China.²²

For diabetes, the AUC values ranged from 0.7524⁴³ to 1.³⁶ The prediction for chronic kidney disease ranged from 0.788¹² to 1.³⁵ For liver-related diseases, an AUC of 0.763 was found for fatty liver disease,⁴⁷ 0.99 for live cancer,⁵¹ and 0.876 for non-alcoholic fatty liver disease.³⁷ Hypertension ranged from 0.766²⁸ to 0.917.²⁴ Other diseases were also evaluated, such as dementia with an AUC of 0.79,⁶⁰ cardiovascular problems with an AUC of 0.74,⁵⁸ and Parkinson with an AUC of 0.97.²⁷ Lung cancer also reached an AUC of 1 by analyzing 153 individuals.²²

The machine learning models with the best performance to predict diseases in the selected studies were K-nearest neighbor, Naive Bayes, deep neural networks, and random forest, all with an AUC of 1. The lowest performing AUC values in the selected studies were support vector machine (AUC of 0.74) and artificial neural network (AUC of 0.7524) (see Table 1).

Risk of bias

Table 2 shows the TRIPOD scale. None of the 30 studies got the maximum score (20 items). In total, 16 studies (53%) received more than 15 items.^{9,26,27,29,34,37–40,43,44,50–52,58,61} Five studies had only nine items,^{22,30,32,36,57} followed by seven studies with 10

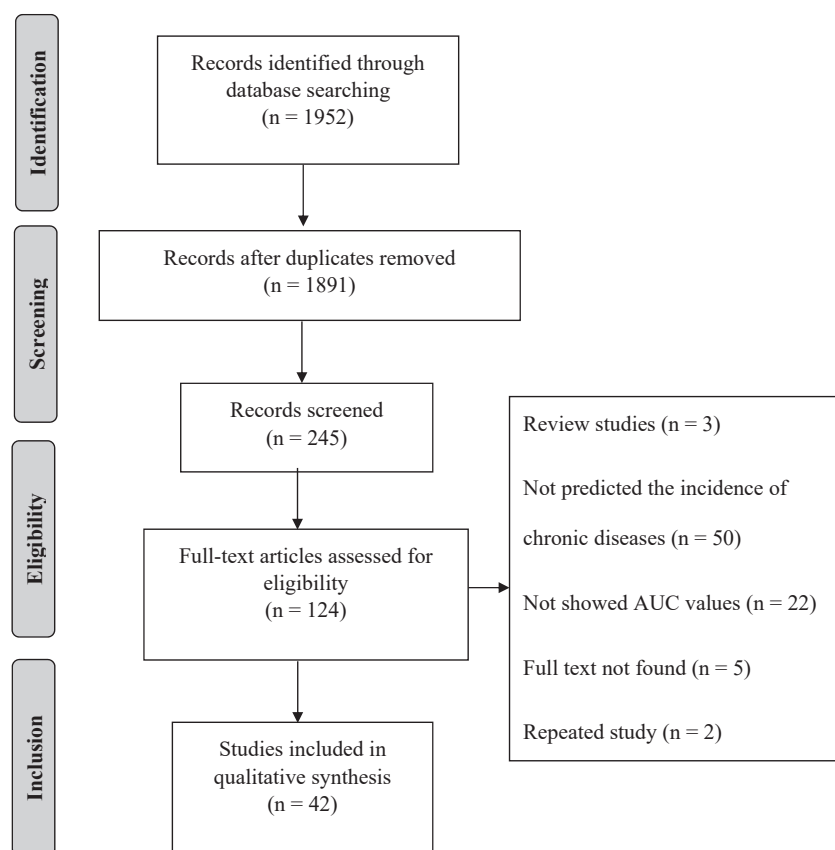


Fig. 1. Flowchart of the selection of studies presented in the review.

items.^{12,31,33,41,46,49,54} Item 15a (present the full prediction model to allow predictions for individuals) and 15b (Explain how to use the prediction model) were not followed by any of the studies.

Discussion

To the best of our knowledge, this is the first systematic review that compared the performance of machine learning algorithms in predicting chronic diseases with AUC values. Our main findings highlight that machine learning algorithms have the potential to predict chronic diseases with good performance and in different contexts. Sociodemographic, anthropometric, and clinical data were the most used characteristics among the studies. We found significant heterogeneity among the studies, mainly in the sample size and in the characteristics used to predict the diseases.

Some of the studies analyzed few individuals, whereas others used very large databases with more than 100,000 subjects. Making estimates with a low number of individuals can lead to prediction bias and overfitting. In some studies, the number of individuals analyzed may have been too small to assess the real predictive quality of the models. For example, the AUC values of two studies that used 50 and 100 individuals were 1 and 0.997, respectively.^{33,42} In contrast, three studies that used a large number of data obtained AUC values of 0.917 (680,810 individuals),²⁴ 0.876 (108,139 individuals),³⁷ and 0.869 (79,313 individuals).¹⁹ These discrepancies point to the importance of using large databases for more accurate predictions and more reliable estimates.

The use of the TRIPOD guidelines highlighted a lack of important information from the predictive studies. Two items, 15a (present the full prediction model to allow predictions for individuals) and 15b (explain how to use the prediction model), were not present in

any studies, which hinders the practical application of these models. As machine learning is a promising tool to help health professionals predict health outcomes, subsequent studies need to be concerned with adequately communicating their final models. The absence of the aforementioned items decreases the chance of the algorithm's application, making the use by health professionals and health managers virtually non-viable.⁶²

Despite the promising results found in our review, some issues deserve further wariness. In general, predictive results are applicable for specific populations, and their findings cannot be immediately extrapolated to the general population without proper validation. In addition, some interventions based on machine learning can increase health costs in the short term and also lead to overdiagnosis. It is not enough that predict models are implemented; it is also necessary to invest in preventive medicine and to define the adequate interventions that arise from these predictions. The lack of detailing in some items found in our review regarding the availability of the algorithms also deserves caution because this is an essential point in increasing thrust among providers and health professionals regarding the use of these models.⁶²

Driven by the increase of available databases, machine learning presents new opportunities for health care systems.⁶³ It has the potential of improving the decision-making capacity of health care professionals and therefore establishing new standards of effectiveness and efficiency in clinical and management operations.⁶⁴ This is clear when we look at the growth in the number of publications in machine learning. Our results confirm these hypotheses, demonstrating that most of the studies (74%) are from the last 2 years.

There are a few limitations in this review. The first one is that only AUC values were included to assess predictive performance.

Table 1

Detailed description and main results of each study included in the systematic review.

Study	Country	Sample	Input features	Outcome	Models	AUC results and best performing model
Barakat et al., 2010 ⁴⁸	Oman	4682 individuals	Sociodemographic and behavioral: age, sex, family history of diabetes Anthropometric: BMI, waist circumference, hip circumference Clinical data: blood pressure, cholesterol, fasting blood sugar, post glucose load	Diabetes	Support vector machine	0.9489
Amaral et al., 2012 ⁴²	Brazil	50 individuals	Sociodemographic and behavioral: age Anthropometric: weight and height Clinical data: forced expiratory volume in the first second, forced vital capacity	Chronic obstructive pulmonary disease	Linear Bayes Normal Classifier, K-nearest neighbor, decision trees, artificial neural networks, and support vector machine	1 (K-nearest neighbor)
Han et al., 2015 ¹⁹	China	79,313 individuals	Sociodemographic and behavioral: age, gender, family history Anthropometric: BMI Clinical data: glycated hemoglobin, triglycerides, cholesterol, blood pressure	Diabetes	Support vector machine, random forest, C4.5, NBTree, and BP NN	0.869 (random forest)
Hornbrook et al., 2018 ⁴¹	The United States	17,095 individuals	Clinical data: blood count	Colorectal cancer	ColonFlag	0.80
17 Olivera et al., 2017 ⁴³	Brazil	12,447 individuals	Sociodemographic and behavioral: physical activity, alcohol intake, consumption of fruits and vegetables, education, smoker, age, use of antihypertensive, lipid-lowering, and other drugs, sleep quality, marital status, coffee consumption Anthropometric: BMI Clinical data: self-report heart failure, myocardial infarction, revascularization, stroke, pain/discomfort in the legs while walking, hypertension, diabetes, and high cholesterol	Diabetes	Logistic regression, artificial neural network, Naïve Bayes, K-nearest neighbor, and random forest	0.7524 (artificial neural network)
Pekkala et al., 2017 ⁴⁵	Finland	709 individuals	Sociodemographic and behavioral: age Anthropometric: BMI Clinical data: cognition, executive functioning, memory, psychomotor speed, MMSE, verbal expression, blood pressure, APOE genotype, cholesterol	Dementia	Disease State Index	0.79
Aljaaf et al., 2018 ⁴⁹	Iraq	400 individuals	Sociodemographic and behavioral: age Clinical data: glucose, urea, creatinine, sodium, potassium, hemoglobin, packed cell	Chronic kidney disease	Recursive partitioning and regression trees, support vector machine, logistic regression, and multilayer perceptron	0.995 (multilayer perceptron)

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Table 1 (continued)

Study	Country	Sample	Input features	Outcome	Models	AUC results and best performing model
Fialoke et al., 2018 ³⁷	The United States	108,139 individuals	volume, white and red blood cell count, specific gravity, urine glucose, albumin, bacteria, red blood cells in urine, plus cell, hypertension, blood pressure, diabetes, coronary artery disease, appetite, pedal edema, anemia Sociodemographic and behavioral: age, gender, and race Clinical data: diabetes	Non-alcoholic liver steatosis	Logistic regression, decision tree, random forest, and XGBoost	0.876 (XGBoost)
Islam et al., 2018 ⁴⁷	Taiwan	994 individuals	Sociodemographic and behavioral: age, gender Anthropometric: BMI Clinical data: cholesterol, glucose, triglycerides	Fatty liver disease	Random forest, support vector machine, artificial neural network, and logistic regression	0.763 (logistic regression)
Pei et al., 2018 ²⁰	China	1200 individuals	Sociodemographic and behavioral: age, occupation, smoking, alcohol, family history of hypertension, and coronary heart disease Anthropometric: height, weight Clinical data: genetic factors	Hypertension	Support vector machine	0.886
Ye et al., 2018 ²⁴	China	680,810 individuals	Sociodemographic and behavioral: age, gender, and race	Hypertension	XGBoost	0.917
Almasoud et al., 2019 ³²	India	400 individuals	Sociodemographic and behavioral: age Clinical data: blood pressure, specific gravity, albumin, sugar, red blood cells, plus cell, bacteria, blood glucose, blood urea, serum creatinine, sodium, potassium, hemoglobin, packed cell volume, white blood cell count, red blood cell count, hypertension, diabetes, coronary artery disease, appetite, pedal edema, anemia	Chronic kidney disease	Logistic regression, support vector machine, random forest, and gradient boosting	0.99 (support vector machine and gradient boosting)
Basak et al., 2019 ³³	India	100 individuals	Sociodemographic and behavioral: age Clinical data: blood pressure, specific Gravity, sugar, blood Glucose, albumin, white and red blood cell count, bacteria, sodium, potassium, serum creatinine, blood urea, hemoglobin, red blood cell count, coronary artery disease, packed cell, pus cell clumps, pus cell, diabetes, anemia, appetite, and class	Chronic kidney disease	Naive Bayes, In-stance-based learning, random forest, decision stump, and decision tree	0.997 (random forest)
Changala and Rao, 2019 ³⁴	India	768 individuals	Sociodemographic and behavioral: pregnancies, age Anthropometric: BMI, skin thickness Clinical data: glucose, blood pressure, insulin, diabetes	Diabetes	Logistic regression, Linear discriminant analysis, K-nearest neighbors, classification and regression trees, Naive Bayes, and support vector machine	0.993 (support vector machine)

Hathaway et al., 2019 ³⁸	The United States	50 individuals	Clinical data: genetic data	Diabetes	Logistic regression, linear discriminant analysis, Gaussian naïve Bayes, support vector machine, and classification and regression tree	0.8 (linear discriminant analysis)
Kriplani et al., 2019 ³⁵	India	224 individuals	25 unspecified attributes	Chronic kidney disease	Naive Bayes, deep neural network, logistic regression, random forest, AdaBoost, and support vector machine	1 (Naive Bayes, deep neural network, and random forest)
Lai et al., 2019 ⁹	Canada	13,309 individuals	Sociodemographic and behavioral: age, sex Anthropometric: BMI Clinical data: fasting blood glucose, cholesterol, triglycerides, blood pressure	Diabetes	Logistic regression and gradient boosting machine	0.847 (gradient boosting machine)
Li et al., 2019 ²⁵	China	965 individuals	Sociodemographic and behavioral: age, exercise, history of hypertension, alcohol Anthropometric: weight Clinical data: genetic data	Hypertension	Support vector machine	0.817
Li et al., 2019 (2) ²⁶	China	5971 individuals	Clinical data: coronary atherosclerotic heart disease, pulmonary edema, disorder of amino acid metabolism, viral pneumonia, auricular fibrillation, chronic kidney disease, headache, hypertensive heart disease, cardiomyopathy	Stroke and heart failure	Naive Bayes, support vector machine, and logistic regression	0.86 for stroke and 0.93 for heart failure (logistic regression)
Mezzatesta et al., 2019 ⁵⁸	Italy and the United States	1216 individuals	Sociodemographic and behavioral: age, sex, smoking, use of antihypertensive drugs, educational, and marital status Clinical data: diabetes, cardiovascular comorbidities, cholesterol, dialysis vintage, hemoglobin, calcium, phosphate, albumin, C-reactive protein, serum glucose, glycated hemoglobin, malignancies, creatinine, AST, ALT	Cardiovascular death, heart failure, ischemia, arrhythmia, other cardiovascular disease	Support vector machine	0.74
Hall et al., 2019 ⁴⁶	Finland	408 individuals	Sociodemographic and behavioral: education Clinical data: cognition, functioning, APOE genotype	Dementia	Disease state index	0.73 for dementia and 0.68 for Alzheimer's
Taninaga et al., 2019 ⁵³	Japan	25,942 individuals	Clinical data: <i>Helicobacter pylori</i> infection status, endoscopic findings, and blood test results	Gastric cancer	XGBoost	0.899
Ullah et al., 2019 ⁵⁰	Pakistan	202 individuals	Sociodemographic and behavioral: age, gender, lifestyle Clinical data: other allergic diseases, asthma, wheezing episodes	Asthma	Artificial neural network, support vector machine, and random forest	0.965 (neural network)

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Table 1 (continued)

Study	Country	Sample	Input features	Outcome	Models	AUC results and best performing model
Xie et al., 2019 ³⁹	The United States	138,146 individuals	Sociodemographic and behavioral: general health, age, doctor visits, income, race, area of residence, employment, sex, marital status, education, sleep time, primary source health care coverage, physical activity, own or rent home, exercises, smoking Anthropometric: BMI Clinical data: flu, angina, coronary heart disease, stress, depression, kidney disease, health problem with special equipment, vision problem, difficulty concentrating, health care coverage, diabetes	Diabetes	Support vector machine, decision tree, logistic regression, random forest, neural network, and Gaussian naive bayes classifiers	0.7949 (neural network)
Xiong et al., 2019 ³⁰	China	11,845 individuals	Sociodemographic and behavioral: gender, age Anthropometric: BMI Clinical data: blood pressure, glycated hemoglobin, triglycerides, cholesterol, and glucose	Diabetes	Multilayer perceptron, AdaBoost, Trees random forest, Support vector machine, and Gradient tree boosting	0.97 (combination of the five models)
Boeri et al., 2020 ⁵⁴	Italy	1021 individuals	Sociodemographic and behavioral: gender, menopause Anthropometric: BMI Clinical data: genetic and previous breast cancer	Breast cancer	Artificial neural network and Support vector machine	0.916 (artificial neural network)
Cao et al., 2020 ²⁷	China	120 individuals	Clinical data: 46 resting-state functional connectivity features, 1 mean amplitude of low-frequency fluctuation feature, 3 mean regional homogeneity features, 1 voxel mirrored homotopic connectivity feature, 2G gray matter volume features, and 1 clinical factor	Parkinson	Random forest and Support vector machine	0.97 (Support vector machine)
Hashem et al., 2020 ⁵¹	Egypt	4423 individuals	Sociodemographic and behavioral: age, gender, smoking Clinical data: AST, ALT, ALP, platelets, AFP, albumin, total bilirubin, hemoglobin, creatinine, WBC	Liver cancer	Classification and regression tree, Alternating decision tree, reduce pruning error tree, and Linear regression algorithm	0.99 (Alternating decision tree)
Huang et al., 2020 ⁵⁵	Germany	2142 individuals	Sociodemographic and behavioral: age, sex, smoking, use of lipid lowering drugs, antihypertensive and anti-diabetic medication Anthropometric: BMI Clinical data: 125 target metabolites, systolic blood pressure, triglyceride, total cholesterol, HDL cholesterol, fasting glucose, eGFR, and UACR	Chronic kidney disease	Support vector machine, random forest, and adaptive boosting	0.857 (support vector machine)

Ma et al., 2020 ²⁹	China	633 individuals	Sociodemographic and behavioral: age, sex, smoking Anthropometric: BMI Clinical data: 9 SNPs features	Chronic obstructive pulmonary disease	Logistic regression, Artificial neural network of the multilayer perceptron, decision tree, XGboost, support vector machine, and K-nearest neighbors	0.94 (XGboost)
Maniruzzaman et al., 2020 ⁴⁰	United States	6,561 individuals	Sociodemographic and behavioral: age, gender, race, education, marital status, occupation, and physical activity Anthropometric: weight, height, BMI Clinical data: blood pressure, cholesterol	Diabetes	Naïve Bayes, Decision tree, Adaboost, and random forest	0.95 (random forest)
Roth et al., 2020 ⁵²	Switzerland	12,761 individuals	Sociodemographic and behavioral: birth year Anthropometric: hip maximum circumference Clinical data: GFR 2, GFR median, no anal cancer max, GFR 1, no anal cancer 2, tenofovir disoproxil fumarate 2, creatinine, hemoglobin, platelet count, CD4 cell count, alkaline phosphatase, serum glucose, HDL	Chronic kidney disease	Elastic net regularization, random forest, gradient boosting machine, multilayer perceptron, and recurrent neural networks	0.996 (Gradient boosting)
Shih et al., 2020 ¹²	Taiwan	19,270 individuals	Sociodemographic and behavioral: gender, age Clinical data: red blood cells, glucose fasting, triglycerides, total cholesterol, HDL and LDL, albumin, proteinuria, urine protein and creatinine ratio, glomerular filtration ratio	Chronic kidney disease	Classification and regression tree, C4.5 decision tree, Linear discriminant analysis, and extreme learning machine	0.788 (C4.5 decision tree)
Tennenhouse et al., 2020 ⁴⁴	Canada	637 individuals	Clinical data: depressive and anxiety instruments	Depression and anxiety	Logistic regression, neural network, and random forest	0.91 for depression and 0.83 for anxiety (logistic regression)
Tigga et al., 2020 ³⁶	India	952 individuals	Sociodemographic and behavioral: age, gender, family history with diabetes, physical activity, smoking, alcohol, hours of sleep, hours of sound sleep, regular intake of medicine, junk food consumption, stress, number of pregnancies Anthropometric: BMI Clinical data: blood pressure, blood pressure, gestation diabetes, frequency of urination, diabetes	Diabetes	Logistic regression, K-Nearest neighbor, Support vector machine, Naive bayes, decision tree, and random forest	1 (random forest)
Zhang et al., 2020 ²⁸	China	625 individuals	Sociodemographic and behavioral: age, gender, alcohol, salty taste, smoking Anthropometric: BMI Clinical data: hyperglycemia, hypertension, dyslipidemia, WHCT, HCT, MCHC, T-BIL, D-BIL	Hypertension, hyperglycemia, and dyslipidemia	Neural network	0.880 for hyperglycemia, 0.766 for hypertension, and 0.703 for dyslipidemia

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Table 1 (continued)

Study	Country	Sample	Input features	Outcome	Models	AUC results and best performing model
Zhang et al., 2020 (1) ⁵⁶	Australia	236,684 individuals	Sociodemographic and behavioral: age, health insurance, income, family history of diabetes, education, alcohol, remoteness, gender, ethnicity Anthropometric: BMI Clinical data: hypertension, cardiovascular disease, dyslipidemia, SES, psychological distress	Diabetes	Random forest, multilayer feedforward, artificial neural network, and gradient boosting machine	0.7927 (gradient boosting machine)
Zhang et al., 2020 (2) ³¹	China	36,652 individuals	Sociodemographic and behavioral: age, sex, education, marital status, income, high fat diet, sweet flavor, history of diabetes Anthropometric: waist circumference, BMI, waist to hip ratio Clinical data: pulse pressure, heart rate, cholesterol, triglyceride, insulin, creatinine, uric acid, urinary protein, hypertension, coronary heart disease	Diabetes	Logistic regression, classification and regression tree, artificial neural networks, support vector machine, random forest, and gradient boosting machine	0.872 (Gradient boosting machine)
Zhu et al., 2020 ²¹	China	709 individuals	Sociodemographic and behavioral: gender, age Clinical data: NLR, hemoglobin, albumin, Carcinoembryonic antigen, CA125, CA199	Gastric cancer	Gradient boosting decision tree	0.91
Al-Azzam et al., 2021 ⁵⁷	Jordan	569 individuals	Clinical data: Wisconsin Diagnosis Cancer dataset	Breast cancer	Logistic regression, Gaussian naïve bayes, Linear support vector machine, support vector machine, decision tree, random forest, XGboost, Gradient boosting, and K-nearest neighbors	0.98 (K-nearest neighbors and gradient boosting machine)
Xie et al., 2021 ²²	China	153 individuals	Clinical data: levels of 61 plasma metabolites	Lung cancer	K-nearest neighbor, Naïve bayes, AdaBoost, support vector machine, random forest, and neural network	1 (K-nearest neighbor, Naïve Bayes, support vector machine, random forest, and neural network)
Wang et al., 2021 ²³	China	5712 individuals	Sociodemographic and behavioral: age, parental history of diabetes, physical activity, history of dyslipidemia Anthropometric: BMI, waist circumference Clinical data: fasting blood glucose, cholesterol, triglycerides, hypertension, blood pressure	Diabetes	Cox proportional hazards, artificial neural network, random forest, and gradient boosting machine	0.851 (Gradient boosting machine)

Table 2

Quality assessment for each study.

	Methods														Results						Total
	Source of data		Participants		Outcome		Predictors		Sample size	Missing data	Analysis methods		Risk groups	Participants	Model development		Model specification				
Reference (Included studies)	4a	4b	5a	5b	6a	6b	7a	7b	8	9	10a	10b	10d	11	13a	13b	14a	14b	15a	15b	
Barakat et al., 2010 ⁴⁸	Yes	No	Yes	Yes	Yes	Yes	Yes	No	No	Yes	Yes	Yes	Yes	Yes	No	No	Yes	Yes	No	No	13
Amaral et al., 2012 ⁴²	Yes	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes	No	Yes	No	No	No	No	13
Han et al., 2015 ¹⁹	Yes	No	Yes	Yes	Yes	No	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes	Yes	Yes	No	No	No	No	13
Hornbrook et al., 2018 ⁴¹	No	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	No	No	Yes	No	No	No	No	No	No	10
Olivera et al., 2017 ⁴³	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes	No	No	No	15
Pekkala et al., 2017 ⁴⁵	No	No	Yes	Yes	No	No	No	Yes	Yes	Yes	Yes	No	No	Yes	Yes	Yes	Yes	Yes	No	No	11
Aljaaf et al., 2018 ⁴⁹	No	No	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes	No	No	No	No	No	No	No	10
Fialoke et al., 2018 ³⁷	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	No	No	No	16
Islam et al., 2018 ⁴⁷	No	No	No	Yes	Yes	No	No	No	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes	Yes	Yes	No	No	11
Pei et al., 2018 ²⁰	Yes	No	No	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	No	No	Yes	Yes	Yes	Yes	No	No	13
Ye et al., 2018 ²⁴	Yes	Yes	Yes	Yes	Yes	No	Yes	No	Yes	No	Yes	Yes	No	No	Yes	Yes	Yes	No	No	No	12
Almasoud et al., 2019 ³²	Yes	Yes	No	No	No	No	Yes	Yes	No	Yes	Yes	Yes	Yes	Yes	No	No	No	No	No	No	9
Basak et al., 2019 ³³	No	No	No	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	No	No	No	No	No	10
Changala and Rao, 2019 ³⁴	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	No	Yes	Yes	No	Yes	Yes	Yes	No	No	15
Hathaway et al., 2019 ³⁸	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes	Yes	No	No	No	16
Kriplani et al., 2019 ³⁵	No	Yes	Yes	Yes	No	No	No	Yes	Yes	Yes	Yes	No	No	Yes	Yes	Yes	Yes	Yes	No	No	12
Lai et al., 2019 ⁹	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	No	Yes	Yes	Yes	Yes	No	No	16
Li et al., 2019 ²⁵	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	No	No	No	15
Li et al., 2019 (2) ²⁶	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes	No	No	16
Mezzatesta et al., 2019 ⁵⁸	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes	No	No	16
Hall et al., 2019 ⁴⁶	No	No	No	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	No	No	No	No	No	10
Taninaga et al., 2019 ⁵³	Yes	Yes	Yes	Yes	Yes	No	Yes	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	No	No	No	14
Ullah et al., 2019 ⁵⁰	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes	No	No	No	15
Xie et al., 2019 ³⁹	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	No	No	No	15
Xiong et al., 2019 ³⁰	Yes	No	Yes	No	Yes	No	Yes	No	Yes	Yes	Yes	Yes	Yes	No	No	No	No	No	No	No	9
Boeri et al., 2020 ⁵⁴	No	Yes	Yes	No	Yes	No	No	No	Yes	Yes	Yes	Yes	Yes	No	No	Yes	Yes	No	No	No	10
Cao et al., 2020 ²⁷	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	No	No	No	15
Hashem et al., 2020 ⁵¹	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes	No	No	No	15
Huang et al., 2020 ⁵⁵	Yes	Yes	No	Yes	Yes	No	Yes	No	No	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes	No	No	13
Ma et al., 2020 ²⁹	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	No	No	No	15
Maniruzzaman et al., 2020 ⁴⁰	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	No	No	No	15
Roth et al., 2020 ⁵²	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes	No	No	No	15
Shih et al., 2020 ¹²	No	No	No	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	No	No	No	No	No	10
Tennenhouse et al., 2020 ⁴⁴	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	No	No	No	15
Tigga et al., 2020 ³⁶	No	No	No	No	Yes	No	Yes	No	Yes	No	Yes	Yes	Yes	Yes	No	No	Yes	Yes	No	No	9
Zhang et al., 2020 ²⁸	Yes	Yes	Yes	Yes	No	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	No	No	No	14
Zhang et al., 2020 (1) ⁵⁶	Yes	Yes	Yes	Yes	Yes	No	Yes	No	Yes	Yes	No	Yes	No	Yes	Yes	Yes	No	No	No	No	12
Zhang et al., 2020 (2) ³¹	Yes	Yes	Yes	Yes	Yes	No	No	No	Yes	Yes	No	Yes	Yes	No	No	Yes	No	No	No	No	10
Zhu et al., 2020 ²¹	Yes	Yes	Yes	Yes	Yes	No	No	No	Yes	No	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes	No	No	13
Al-Azzam et al., 2021 ⁵⁷	Yes	No	Yes	No	No	No	Yes	No	Yes	Yes	Yes	Yes	Yes	Yes	No	No	No	No	No	No	9
Xie et al., 2021 ²²	Yes	No	Yes	No	Yes	No	No	No	Yes	No	Yes	No	Yes	Yes	No	No	Yes	Yes	No	No	9
Wang et al., 2021 ²³	Yes	Yes	Yes	Yes	Yes	No	Yes	No	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes	No	Yes	No	No	14

*No = 0 point, Yes = 1 point. Maximum 20 points.

However, the AUC is the most comprehensive and widely used metric among machine learning studies. A few studies used other metrics, such as precision, accuracy, sensitivity, and specificity. However, as the AUC was the most reported result overall, we decided to use it as the metric. Another limitation is publication bias because only published studies were included, possibly leaving many negative results out that went unpublished.⁶⁵ Finally, it was impossible to perform a meta-analysis of the results because of the significant heterogeneity between the studies, different morbidities, and the insufficient comparable data.

Conclusion

The findings show that machine learning algorithms are a promising tool to predict the occurrence of chronic diseases in different contexts. These algorithms can be helpful for health care professionals and systems to improve clinical decisions, reduce health costs, and promote cost-effectiveness. However, new studies should be conducted in line with the TRIPOD scale recommendations, especially regarding data description, the availability of a practical tool, and an explanation of how to use the predict model. Finally, future studies are expected to use larger databases whenever possible to decrease the risk of overfitting.

Author statements

Ethical approval

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Competing interests

None declared.

Authors' contributions

F.M.D. participated in the design and writing of the article. A.K.C. participated as the second reviewer of the article. S.R.F. participated in the elaboration of the quality scale of the studies. A.D.P.C.F., R.A.A., and B.P.N. participated in the writing and revision of the final version of the article. All authors reviewed and approved the final version to be published, assuming responsibility for all aspects of the work, including ensuring its accuracy and integrity.

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